

Qatar-1b

R-1978

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_130423 · created 2026-08-07T00:29:42+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456405.926118 +/- 2.3e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.185 +/- 0.029
Transit depth (Rp/Rs)^2	3.42 +/- 1.08 [%]
Orbital Inclination (inc)	81.86 +/- 2.2
Airmass coefficient 1 (a1)	0.96 +/- 0.018
Airmass coefficient 2 (a2)	0.029 +/- 0.014
Scatter in the residuals of the lightcurve fit is	1.82 %
Best Comparison Star	#1 - [343, 187]
Optimal Aperture	2.8
Optimal Annulus	15.11
Transit Duration (day)	0.052 +/- 0.023

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.185 ± 0.029 vs 0.1463 (deviation 1.19σ)
Duration fit vs published	0.052 d vs 0.0692 d (deviation 0.66σ)
Mid-time O–C	0.1 min
Depth SNR	5.7
Residual scatter	1.82 %

Light curve

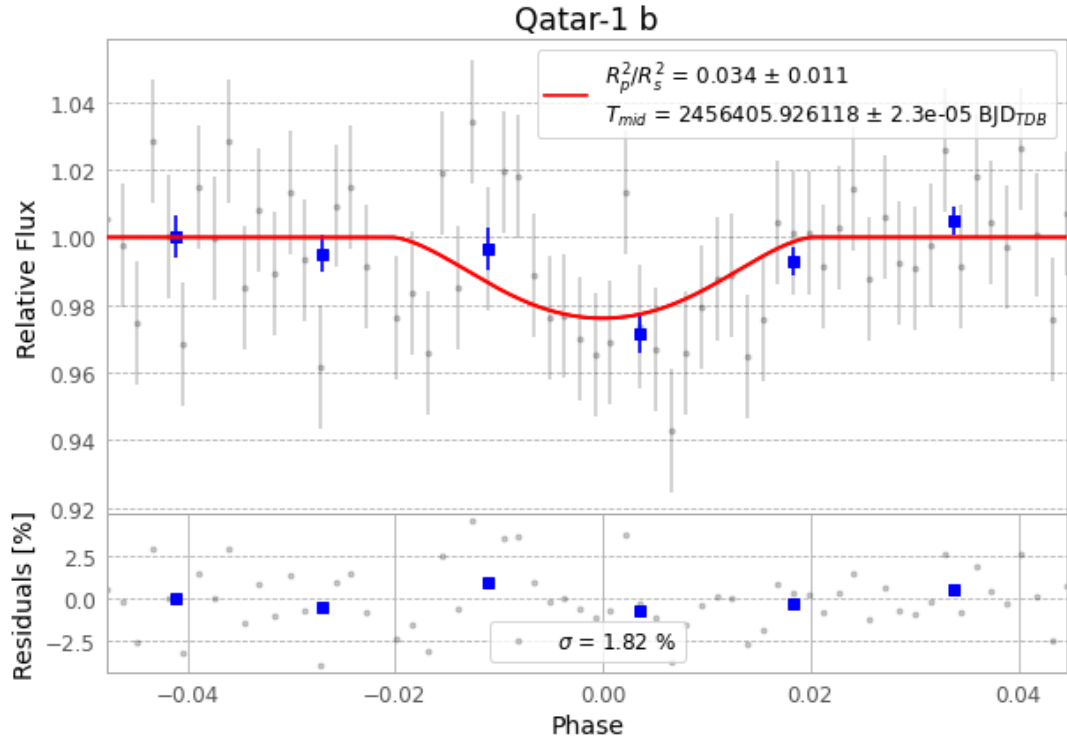


Figure 1 — FinalLightCurve_Qatar-1 b_2013-04-23.png (SHA-256 0009e1de62e37169...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [412, 165], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 0bf91f49210e529c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

64 FITS frames (42 MB), Qatar-1130423083612.FITS → Qatar-1130423114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-130423.log (494811e2ec51...), FinalLightCurve_Qatar-1 b_2013-04-23.png (0009e1de62e3...), FinalLightCurve_Qatar-1 b_2013-04-23.csv (17bd438e7926...)

Compute resources

Wall time 107.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 00:29Z.